TMUA PRACTICE TEST SERIES

TMUA Topic Test 2

Graphs, Trigonometry & Logarithms

- 20 multiple-choice questions
- Recommended time: 75 minutes
- Non-calculator
- Paper 1 styled topic revision set

SOLUTION BOOK

ThrivingScholars

ANSWER KEY

TMUA Topic Test 2 — Graphs, Trigonometry & Logarithms

Complete worked solutions for the supplied 20-question TMUA Paper 1 styled topic test.

Verification complete: all 20 questions, option sets, keyed answers, worked solutions and embedded diagrams have been cross-checked against the interactive test version.

TMUA Topic Test 2 answer key

Q1 **D**	Q2 **A**	Q3 **D**	Q4 **G**	Q5 **C**	Q6 **F**	Q7 **B**	Q8 **F**	Q9 **E**	Q10 **C**
Q11 **B**	Q12 **B**	Q13 **D**	Q14 **B**	Q15 **B**	Q16 **B**	Q17 **E**	Q18 **A**	Q19 **C**	Q20 **D**

WORKED SOLUTIONS

Logarithms & Exponentials

Questions 1–5 · logarithmic equations, identities, parameter reasoning and estimation.

Q1

Logarithms

Answer **D**

Question 1

Find the greatest value of $x+y$, where the real numbers x and y satisfy

$$\log_2\left(\frac{x^2}{y}\right) = 6 \quad \text{and} \quad 12 - \log_2 x = (\log_2 x)(\log_2 y).$$

A $16 + \frac{\sqrt{2}}{2}$

B $32 + \frac{\sqrt{2}}{4}$

C 16

D 20

E 32

F $\frac{1+128\sqrt{2}}{512}$

G 120

H 256

WORKED SOLUTION

Let $a = \log_2 x$ and $b = \log_2 y$. The first equation gives

$$2a - b = 6.$$

The second gives $12 - a = ab$. Substituting $b = 2a - 6$,

$$12 - a = a(2a-6) \iff 2a^2 - 5a - 12 = 0 \iff (2a+3)(a-4) = 0.$$

Thus $(a,b) = (4,2)$ or $(-\frac{3}{2}, -9)$. These produce

$$x + y = 16 + 4 = 20$$

or

$$x + y = 2^{-3/2} + 2^{-9} = \frac{\sqrt{2}}{4} + \frac{1}{512} < 20.$$

Hence the greatest sum is $\boxed{20}$.

The correct answer is D.

Q2

Logarithmic Equations

Answer **A**

Question 2

Let $a > 0$, $a \neq 1$. Find all positive values of $x \neq 1$ satisfying

$$\log_a x + \log_x a = \frac{5}{2}.$$

- **A** $x = a^{1/2}$ or $x = a^2$
- **B** $x = a^{-1/2}$ or $x = a^{-2}$
- **C** only $x = a^{1/2}$
- **D** only $x = a^2$
- **E** $x = a^{-1/2}$ or $x = a^2$
- **F** no such values of x

WORKED SOLUTION

Let

$$t = \log_a x.$$

Because $x \neq 1$, we have $t \neq 0$, and the reciprocal logarithm identity gives

$$\log_x a = \frac{1}{t}.$$

Hence

$$t + \frac{1}{t} = \frac{5}{2}.$$

Multiplying by $2t$,

$$2t^2 - 5t + 2 = 0 \iff (2t-1)(t-2) = 0.$$

Thus $t = \frac{1}{2}$ or $t = 2$. Since $t = \log_a x$,

$$\boxed{x = a^{1/2} \text{ or } x = a^2}.$$

The correct answer is A.

Q3

Logarithms

Answer D

Question 3

You are given

$$\log_a(xy^2z) = 6, \qquad \log_a(x^2yz^4) = 9, \qquad \log_a(x^5y^7z^5) = 25,$$

where $x, y, z > 1$. What is the value of a?

A 1

B x

C y

D z

E There is not enough information to determine whether a is any of the other options.

WORKED SOLUTION

Write

$$A = \log_a x, \qquad B = \log_a y, \qquad C = \log_a z.$$

The three equations become

$$A + 2B + C = 6, \qquad 2A + B + 4C = 9, \qquad 5A + 7B + 5C = 25.$$

Three times the first equation plus the second gives

$$5A + 7B + 7C = 27.$$

Subtracting the third equation gives $2C = 2$, so $C = 1$. Hence

$$\log_a z = 1 \implies z = a.$$

Therefore $\boxed{a = z}$.

The correct answer is D.

Q4

Graphs & Logarithms

Answer G

Question 4

The curve $y = ax^2 + 3$ passes through

$$(2, \log_2 p) \quad \text{and} \quad (-1 + \log_2 p, 11),$$

where $p > 0$. Find p.

A $\frac{1}{32}$	B $\frac{1}{16}$
C $\frac{1}{4}$	D 1
E 4	F 16
G 32	

WORKED SOLUTION

Let $L = \log_2 p$. From the first point,

$$L = 4a + 3.$$

From the second point,

$$11 = a(L-1)^2 + 3 \implies a(L-1)^2 = 8.$$

Using $a = (L-3)/4$,

$$(L-3)(L-1)^2 = 32.$$

Let $t = L - 1$. Then

$$t^3 - 2t^2 - 32 = 0 = (t-4)(t^2 + 2t + 8).$$

The quadratic factor has negative discriminant, so the only real solution is $t = 4$. Thus $L = 5$, and

$$p = 2^5 = \boxed{32}.$$

The correct answer is G.

Q5

Logarithms & Estimation

Answer C

Question 5

Which of the following expressions has the smallest value?

$$\begin{array}{rl} \text{(i)} & 4-\log_{10}\pi, \\ \text{(ii)} & 2+\log_{10}\sqrt{\pi}, \\ \text{(iii)} & \sqrt{3+\log_{10}\pi}, \\ \text{(iv)} & \dfrac{5}{\sqrt{\log_{10}\pi}}. \end{array}$$

A $4-\log_{10}\pi$

B $2+\log_{10}\sqrt{\pi}$

C $\sqrt{3+\log_{10}\pi}$

D $\frac{5}{\sqrt{\log_{10}\pi}}$

WORKED SOLUTION

Use $\pi^2 \approx 10$, so $\log_{10}\pi \approx \frac{1}{2}$. Then the four values are approximately

$$\frac{7}{2}, \qquad \frac{9}{4}, \qquad \sqrt{\frac{7}{2}}, \qquad 5\sqrt{2}.$$

Since $\sqrt{\frac{7}{2}} \approx 1.87$, it is the smallest. Therefore the answer is

$$\boxed{\sqrt{3+\log_{10}\pi}}.$$

The correct answer is C.

WORKED SOLUTIONS

Graphs & Graphical Reasoning

Questions 6–14 · quadratic and polynomial graphs, root counting, curve arrangements and graph interpretation.

Q6

Quadratic Graphs

Answer F

Question 6

Let $f(x)$ be quadratic. The graph of $y = f(x)$ passes through $(-1, -3)$ and has vertex $(1, 5)$. Which expression is $f(x)$?

A $-x^2 + 4x + 3$	B $x^2 - 4x + 3$
C $2x^2 - 4x + 3$	D $-2x^2 - 4x + 3$
E $-x^2 - 4x + 3$	F $-2x^2 + 4x + 3$
G $4x - 2x^2$	H $4x - x^2$

WORKED SOLUTION

Use vertex form:

$$f(x) = A(x-1)^2 + 5.$$

Since $f(-1) = -3$,

$$-3 = 4A + 5 \implies A = -2.$$

Therefore

$$f(x) = -2(x-1)^2 + 5 = -2x^2 + 4x + 3.$$

The answer is $\boxed{\text{F}}$.

The correct answer is F.

Q7

Trigonometric Inequalities

Answer B

Question 7

Find the complete set of x, where $-\pi \le x \le \pi$, satisfying

$$(1 - 2\cos 2x)(1 + 2\sin x) \le 0.$$

A $-\pi \le x \le -\frac{5\pi}{6},\ -\frac{\pi}{6} \le x \le \frac{\pi}{6},\ \frac{5\pi}{6} \le x \le \pi$

B $-\pi \le x \le \frac{\pi}{6},\ \frac{5\pi}{6} \le x \le \pi$

C $-\frac{5\pi}{6} \le x \le -\frac{\pi}{6},\ \frac{\pi}{6} \le x \le \frac{5\pi}{6}$

D $-\pi \le x \le -\frac{5\pi}{6},\ -\frac{\pi}{6} \le x \le \frac{\pi}{6},\ x = \frac{\pi}{2},\ \frac{5\pi}{6} \le x \le \pi$

E $-\pi \le x \le -\frac{5\pi}{6},\ x = \frac{\pi}{2},\ \frac{5\pi}{6} \le x \le \pi$

WORKED SOLUTION

Using $\cos 2x = 1 - 2\sin^2 x$,

$$1 - 2\cos 2x = 4\sin^2 x - 1 = (2\sin x - 1)(2\sin x + 1).$$

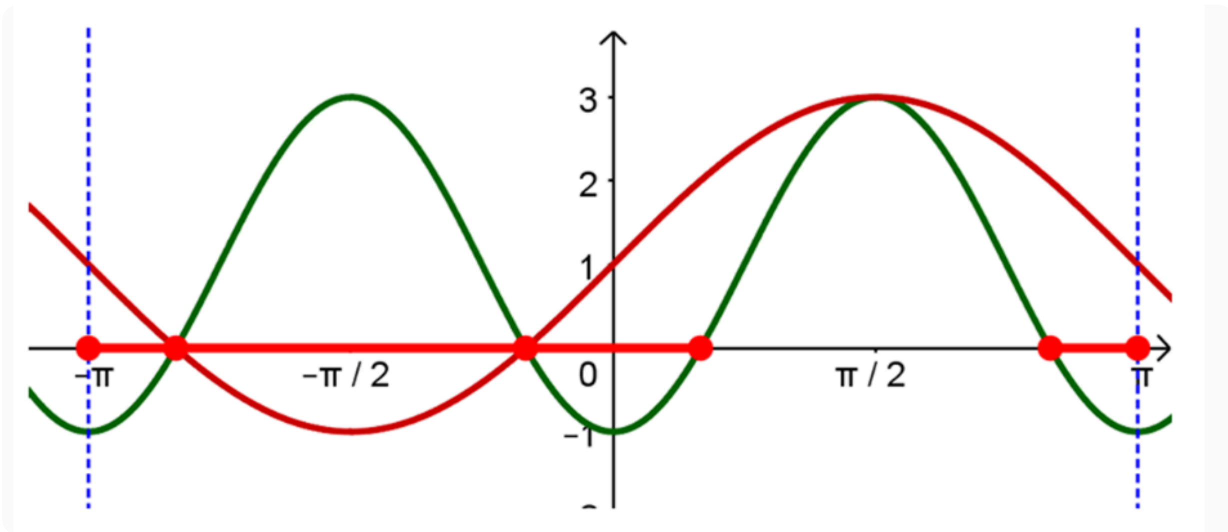
Hence the product is

$$(2\sin x - 1)(2\sin x + 1)^2.$$

The squared factor is non-negative, so the inequality is equivalent to $\sin x \le \frac{1}{2}$. On $[-\pi, \pi]$,

$$-\pi \le x \le \frac{\pi}{6} \quad \text{or} \quad \frac{5\pi}{6} \le x \le \pi.$$

Therefore the answer is $\boxed{\text{B}}$.



The correct answer is B.

Q8

Trigonometric Inequalities

Answer **F**

Question 8

Find the fraction of $0 \le \theta \le 2\pi$ for which

$$\left(2\cos\frac{\theta}{2}+1\right)(\sin\theta+\cos\theta) \le 0.$$

A $\frac{1}{6}$	B $\frac{7}{12}$
C $\frac{3}{4}$	D $\frac{1}{4}$
E $\frac{1}{2}$	F $\frac{5}{12}$
G $\frac{5}{6}$	H $\frac{2}{3}$

WORKED SOLUTION

The first factor changes sign at

$$2\cos\frac{\theta}{2}+1=0 \implies \theta=\frac{4\pi}{3}.$$

Also

$$\sin\theta+\cos\theta=\sqrt{2}\sin\left(\theta+\frac{\pi}{4}\right),$$

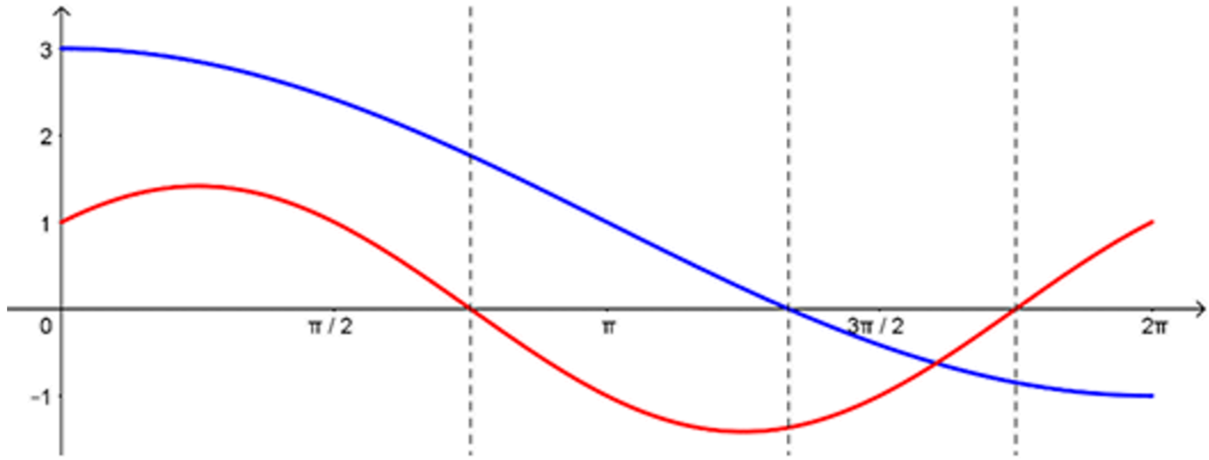
which is zero at $\theta=\frac{3\pi}{4}$ and $\frac{7\pi}{4}$. A sign chart gives

$$\frac{3\pi}{4}\le\theta\le\frac{4\pi}{3} \quad \text{or} \quad \frac{7\pi}{4}\le\theta\le 2\pi.$$

The total length is

$$\left(\frac{4\pi}{3}-\frac{3\pi}{4}\right)+\left(2\pi-\frac{7\pi}{4}\right)=\frac{5\pi}{6}.$$

As a fraction of 2π, this is $\boxed{\frac{5}{12}}$.



The correct answer is F.

Q9

Trigonometric Equations & Graphs

Answer E

Question 9

Find the number of solutions of

$$x \tan x = \sin x, \qquad -\pi \le x \le \pi.$$

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

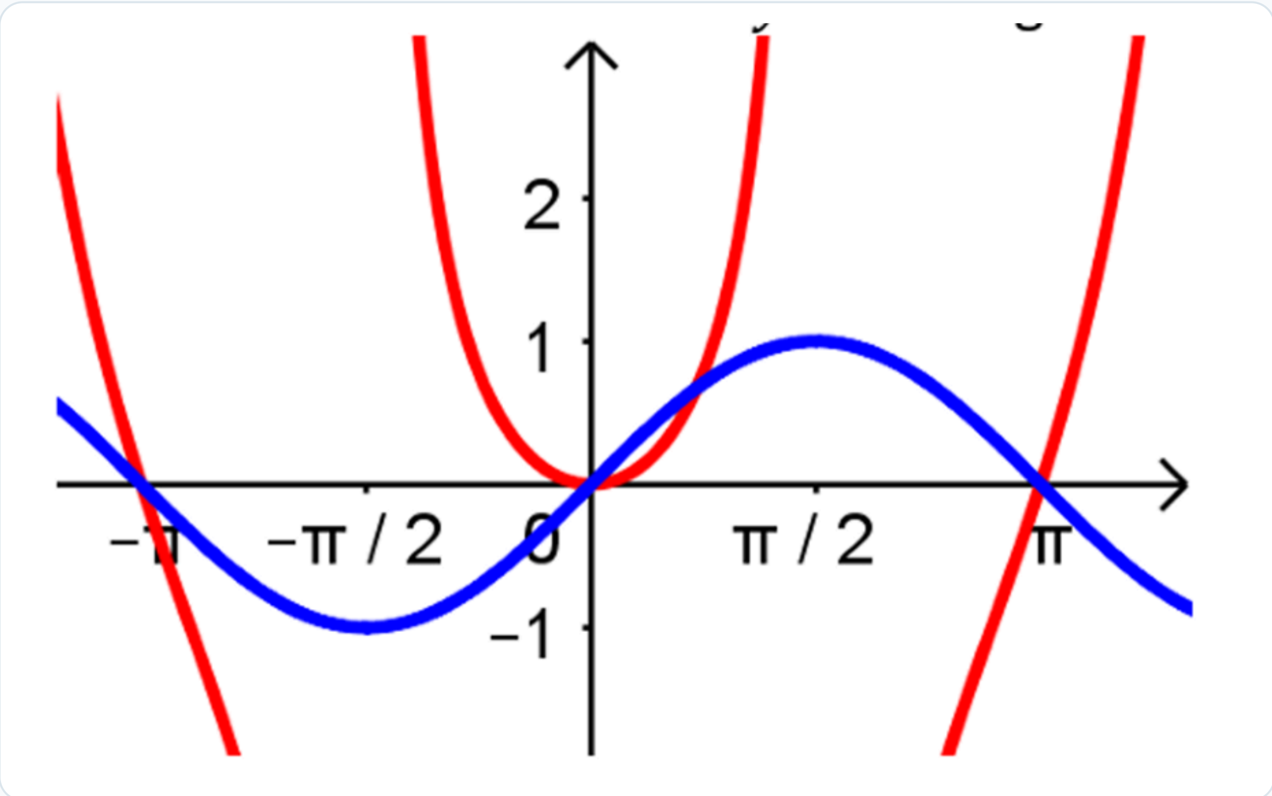
At points where the equation is defined,

$$x \tan x = \sin x \iff \sin x \left(\frac{x}{\cos x} - 1 \right) = 0.$$

First, $\sin x = 0$ gives $x = -\pi, 0, \pi$, all of which are valid.

Otherwise, $x = \cos x$. Since $x - \cos x$ is increasing on $[-1, 1]$, this equation has exactly one solution there. Hence the total number is

$$3 + 1 = \boxed{4}.$$



The correct answer is E.

Q10

Polynomial Graphs

Answer C

Question 10

How many real roots does

$$3x^5 - 10x^3 - 225x + 10 = 0$$

have?

A 1

B 2

C 3

D 4

E 5

WORKED SOLUTION

Let $P(x) = 3x^5 - 10x^3 - 225x + 10$. Then

$$P'(x) = 15(x^4 - 2x^2 - 15) = 15(x^2 - 5)(x^2 + 3).$$

The only real stationary points are $x = \pm\sqrt{5}$. Moreover,

$$P(-\sqrt{5}) = 10 + 200\sqrt{5} > 0, \qquad P(\sqrt{5}) = 10 - 200\sqrt{5} < 0.$$

Since $P(x) \to -\infty$ as $x \to -\infty$ and $P(x) \to \infty$ as $x \to \infty$, the graph crosses the axis once before the local maximum, once between the two stationary points, and once after the local minimum.

Therefore there are $\boxed{3}$ real roots.

The correct answer is C.

Q11

Curve Intersections

Answer **B**

Question 11

Into how many regions is the plane divided by the three curves

$$y = x^3 - x^2, \qquad y = x^3 - 4x, \qquad y = x^2?$$

A	11	B	10
C	9	D	8
E	7		

WORKED SOLUTION

Begin with $y = x^3 - x^2$. A single continuous graph divides the plane into two regions.

Now compare it with $y = x^2$:

$$x^3 - x^2 = x^2 \iff x^2(x-2) = 0.$$

The curves are tangent at $(0,0)$ and cross at $(2,4)$. These two intersection points split the newly added parabola into three open arcs. Tracing the parabola from left to right, each arc lies inside an existing region and divides that region into two. Therefore the first two curves form

$$2 + 3 = 5$$

regions.

The third curve, $y = x^3 - 4x$, meets $y = x^3 - x^2$ when

$$x(x-4) = 0,$$

and it meets $y = x^2$ when

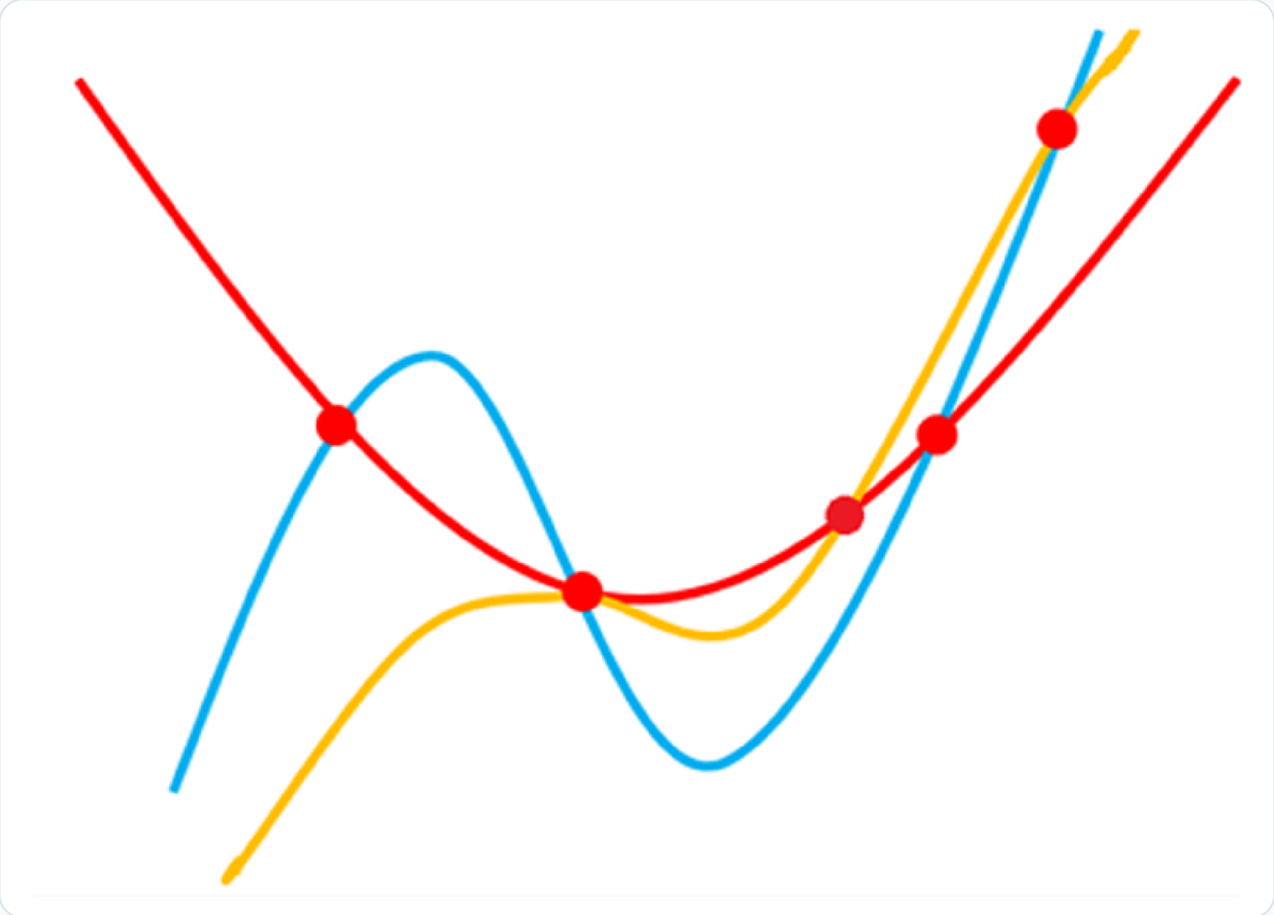
$$x(x^2 - x - 4) = 0.$$

Thus its distinct intersection abscissae with the existing union are

$$x = \frac{1-\sqrt{17}}{2}, \quad 0, \quad \frac{1+\sqrt{17}}{2}, \quad 4.$$

These four points split the third cubic into five arcs. Between consecutive intersection points, its vertical order relative to the other two curves is fixed, so every arc lies in the interior of one existing region and splits it. Hence five more regions are created:

$$5 + 5 = \boxed{10}.$$



The correct answer is B.

Q12

Cubic Graphs

Answer B

Question 12

The equation

$$x^3 + 6x^2 - 63x + k = 0$$

has three **distinct** real solutions when:

A $-108 < k < 392$

B $-392 < k < 108$

C $108 < k < 392$

D $-392 < k < 0$

E $0 < k < 108$

WORKED SOLUTION

Let

$$f(x) = x^3 + 6x^2 - 63x + k.$$

Then

$$f'(x) = 3x^2 + 12x - 63 = 3(x+7)(x-3).$$

Hence the local maximum is at $x = -7$, and the local minimum is at $x = 3$. Their values are

$$f(-7) = k + 392, \qquad f(3) = k - 108.$$

For the cubic to cross the axis three times at distinct points, its local maximum must lie above the axis and its local minimum below the axis:

$$k + 392 > 0, \qquad k - 108 < 0.$$

Therefore

$$\boxed{-392 < k < 108}.$$

The correct answer is B.

Q13

Function Graphs

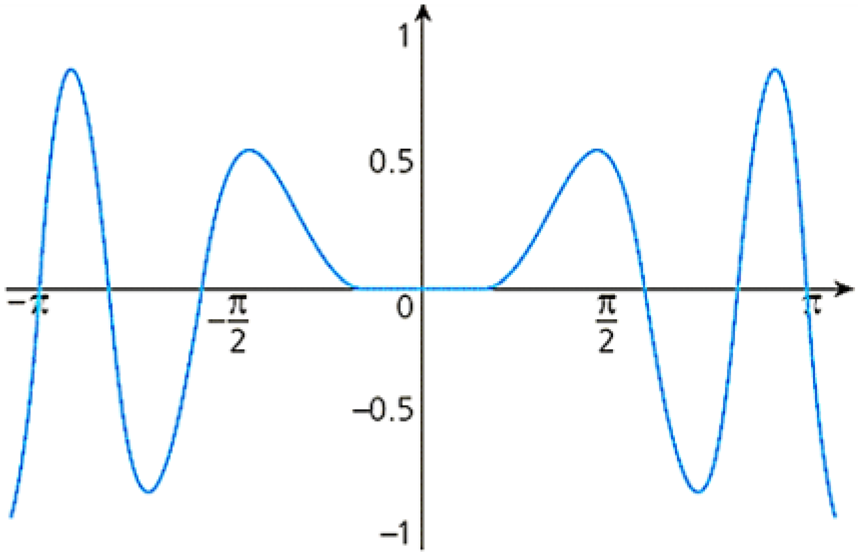
Answer D

Question 13

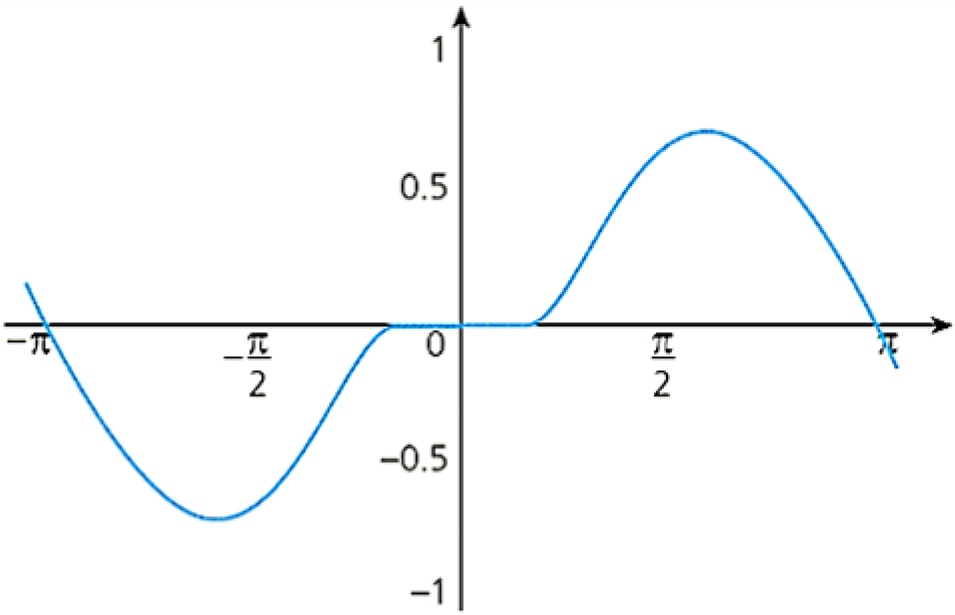
Which graph shows the function

$$f(x) = \begin{cases} e^{-1/x^2}\cos(x^2), & x \neq 0, \\ 0, & x = 0, \end{cases} \qquad -\pi \leq x \leq \pi?$$

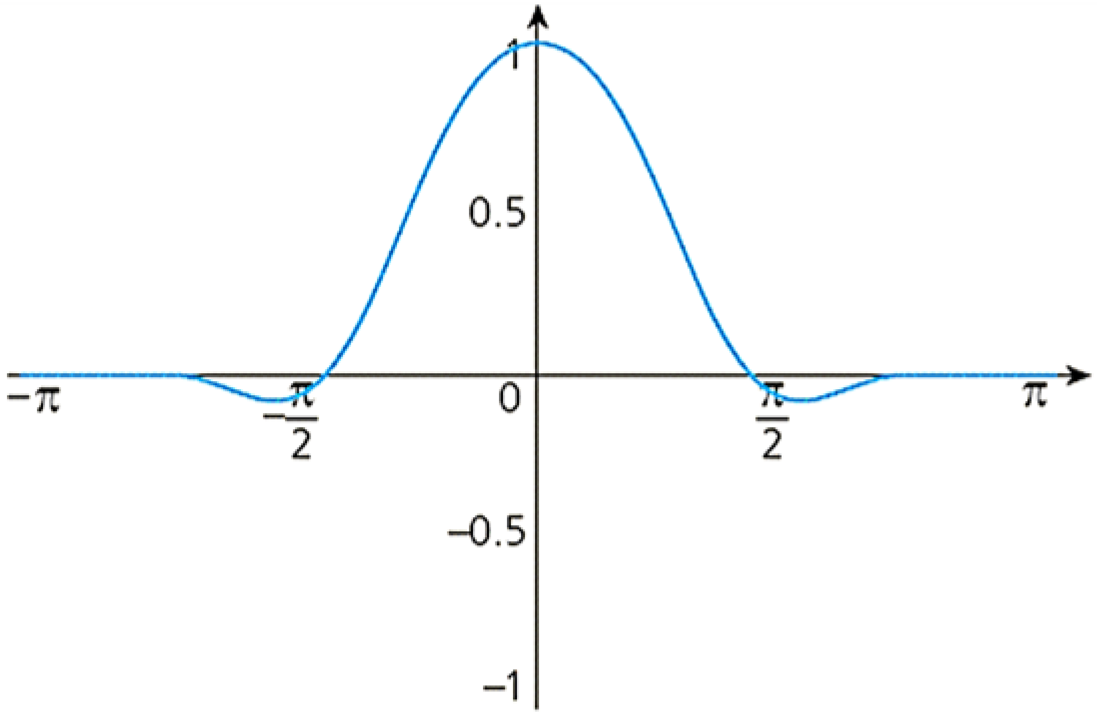
A



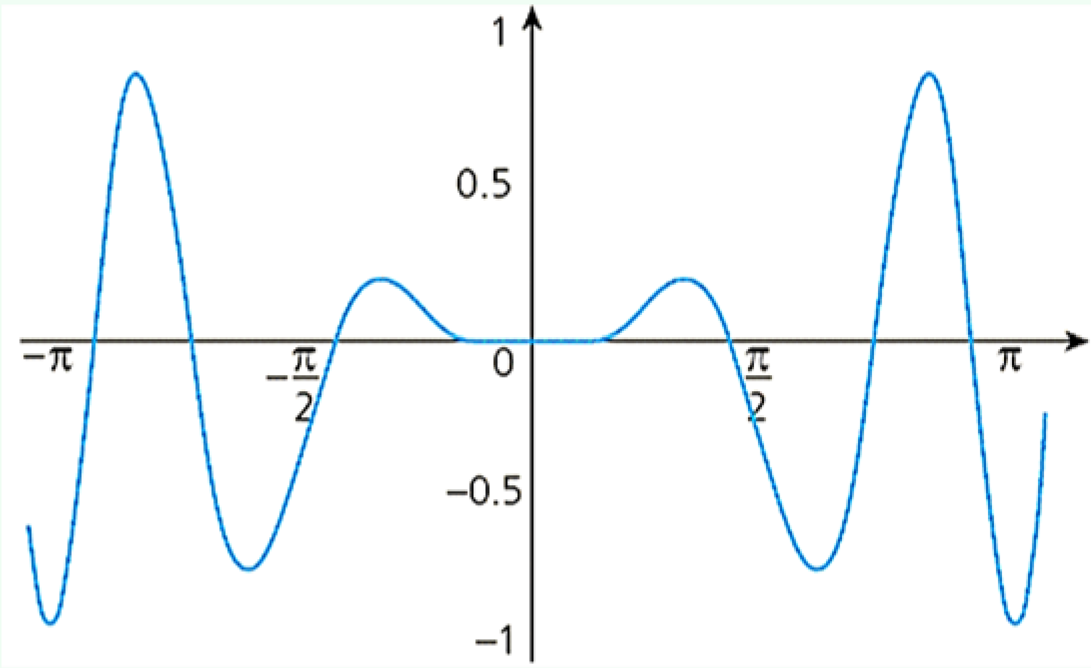
B



C



D



WORKED SOLUTION

The function is even because it depends only on x^2, so its graph is symmetric about the y-axis.

As $x \to 0$, $e^{-1/x^2} \to 0$, and the definition $f(0) = 0$ makes the graph continuous at the origin. The factor e^{-1/x^2} increases with $|x|$, so oscillations have very small amplitude near 0 and larger amplitude farther away.

The zeros away from the origin occur when $\cos(x^2) = 0$:

$$x^2 = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$$

within the given interval. The alternating signs, even symmetry and increasing envelope match only graph $\boxed{\text{D}}$.

The correct answer is D.

Q14

Trigonometric Graphs

Answer **B**

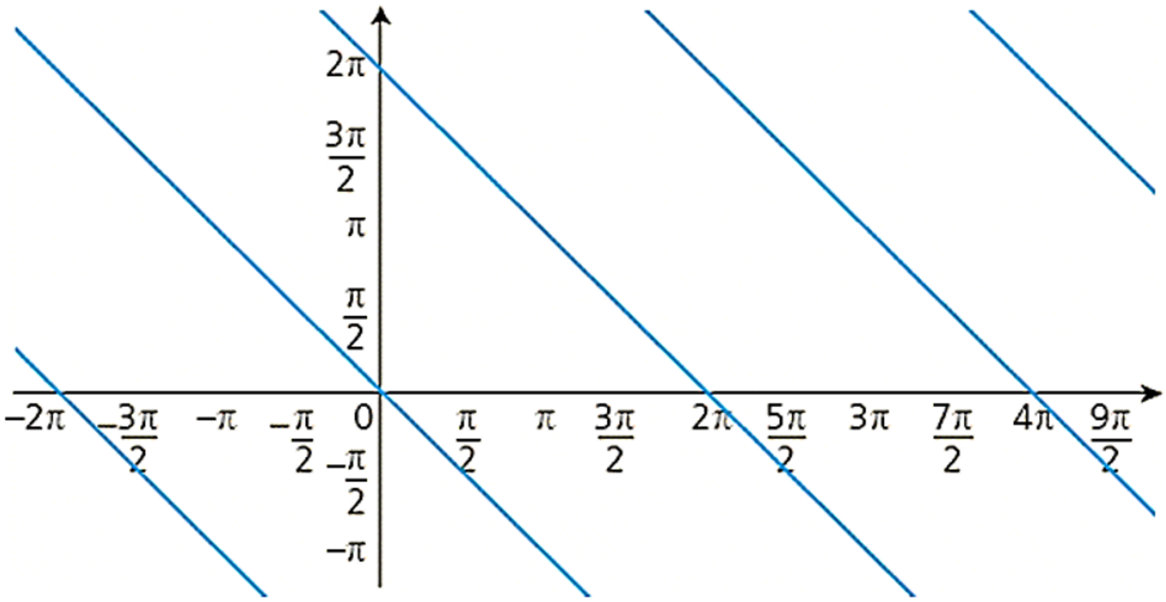
Question 14

The graph of all points (x, y) satisfying

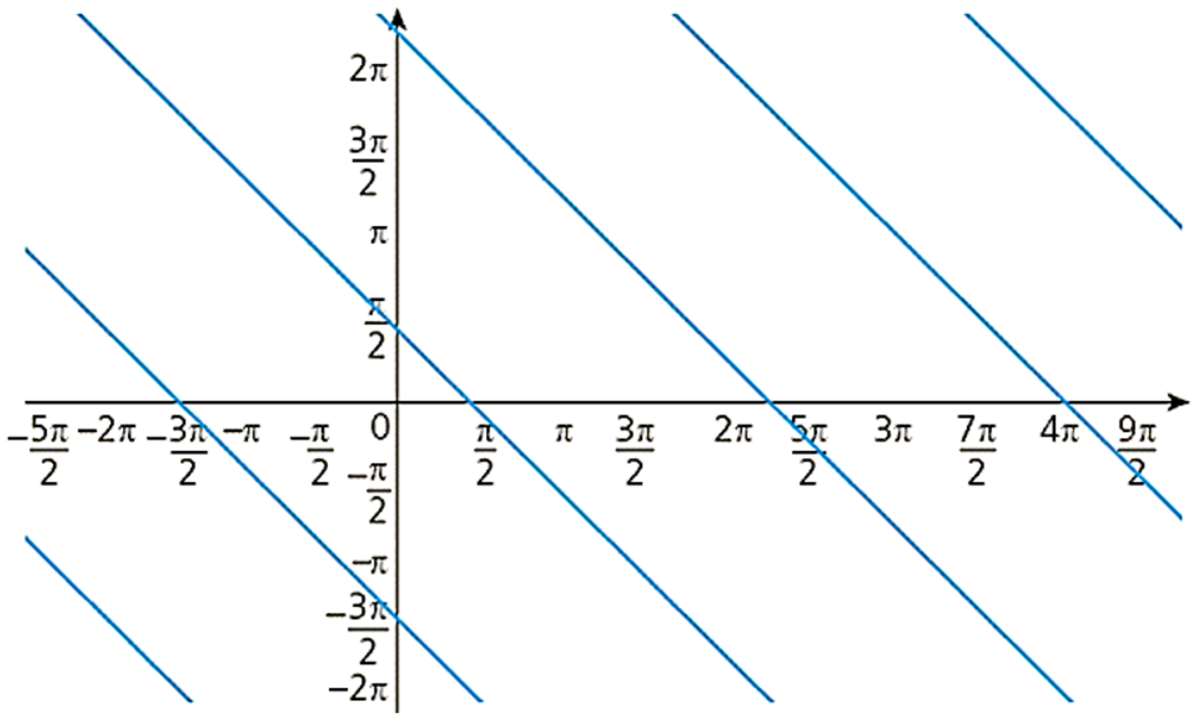
$$\sin(x + y) = 1$$

is shown in:

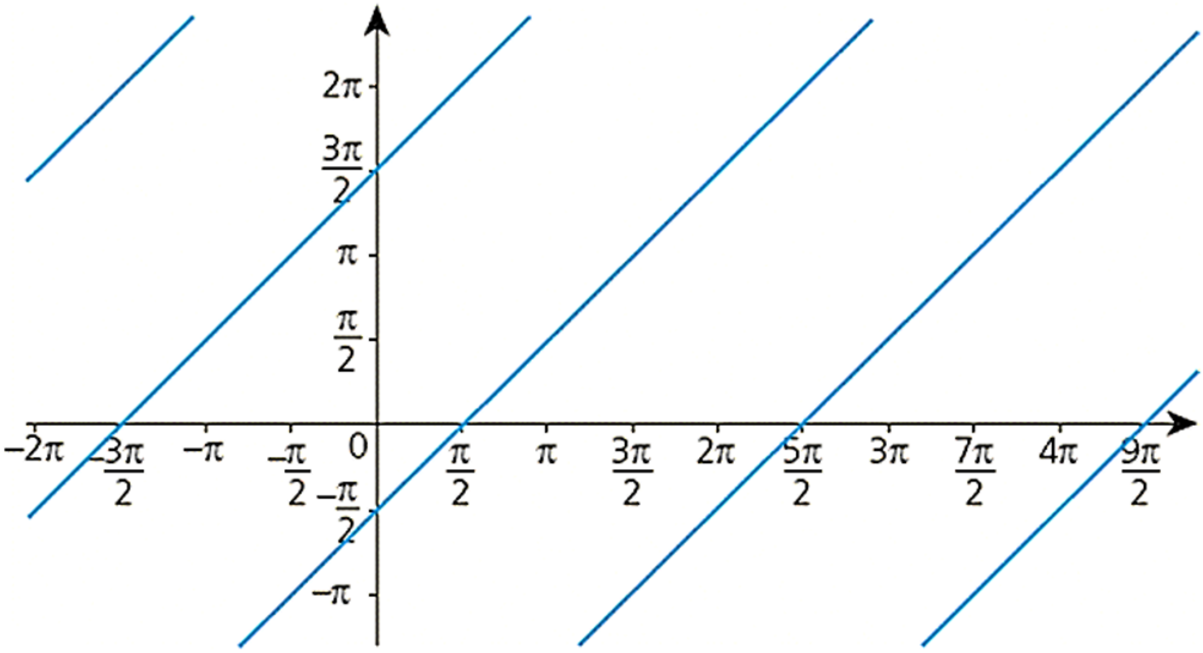
A



B

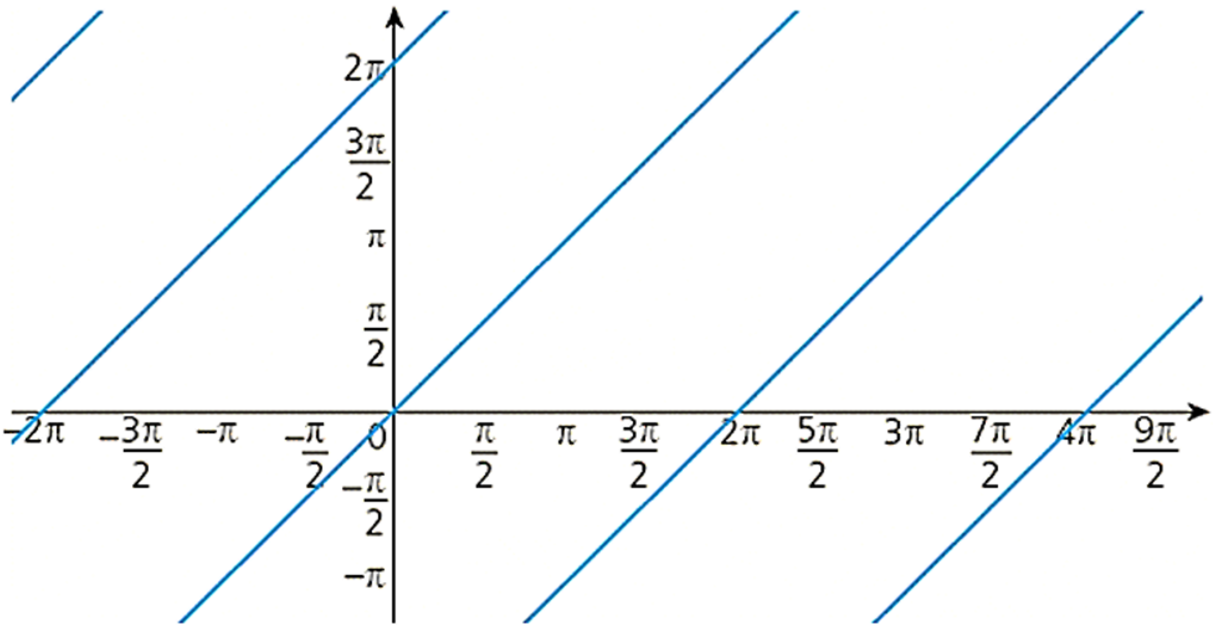


C



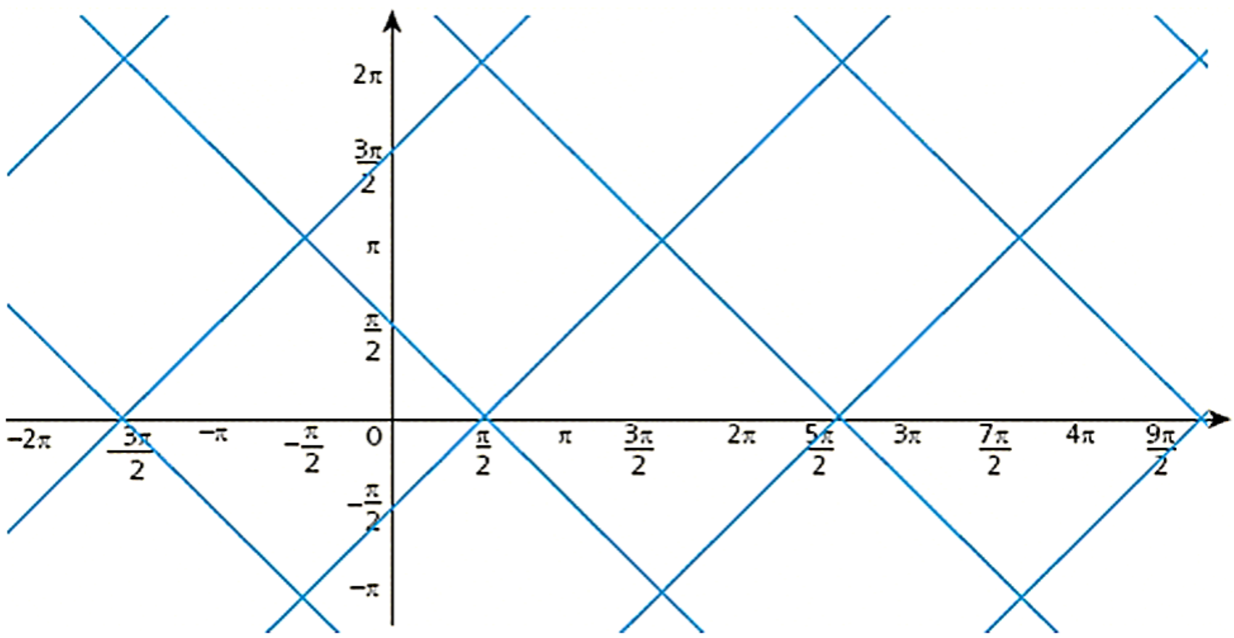
2π
3π/2
π
π/2
−2π
−3π/2
−π
−π/2
0
−π/2
π/2
π
3π/2
2π
5π/2
3π
7π/2
4π
9π/2
−π

D



2π
3π/2
π
π/2
−2π
−3π/2
−π
−π/2
0
−π/2
π/2
π
3π/2
2π
5π/2
3π
7π/2
4π
9π/2
−π

E



2π
3π/2
π
π/2
−2π
3π/2
−π
−π/2
0
−π/2
π/2
π
3π/2
2π
5π/2
3π
7π/2
4π
9π/2
−π

WORKED SOLUTION

The sine function equals 1 precisely when

$$x + y = \frac{\pi}{2} + 2k\pi, \qquad k \in \mathbb{Z}.$$

Therefore

$$y = -x + \frac{\pi}{2} + 2k\pi.$$

This is a family of equally spaced parallel lines, each with slope -1. The correct diagram is $\boxed{\text{B}}$.

The correct answer is B.

WORKED SOLUTIONS

Trigonometry

Questions 15–20 · inequalities, sequences, equations, geometry and maximum-value reasoning.

Q15

Trigonometric Inequalities

Answer **B**

Question 15

Find the fraction of $0 \le \theta \le 2\pi$ for which

$$(\cos\theta + \sin\theta)\left(\frac{\sqrt{3}}{2} + \cos\frac{\theta}{2}\right)(\cos^2\theta - 1) \le 0.$$

- **A** $\frac{1}{4}$
- **B** $\frac{5}{12}$
- **C** $\frac{1}{2}$
- **D** $\frac{2}{3}$
- **E** $\frac{3}{4}$
- **F** $\frac{5}{6}$

WORKED SOLUTION

Since

$$\cos^2\theta - 1 = -\sin^2\theta \le 0,$$

the inequality is satisfied, apart from isolated equality points, when the first two factors have the same sign.

Now $\cos\theta + \sin\theta \ge 0$ on

$$0 \le \theta \le \frac{3\pi}{4} \quad \text{and} \quad \frac{7\pi}{4} \le \theta \le 2\pi,$$

while

$$\frac{\sqrt{3}}{2} + \cos\frac{\theta}{2} \ge 0 \quad \text{for} \quad 0 \le \theta \le \frac{5\pi}{3}.$$

The factors have the same sign on

$$0 \le \theta \le \frac{3\pi}{4} \quad \text{and} \quad \frac{5\pi}{3} \le \theta \le \frac{7\pi}{4}.$$

The total length is $\frac{5\pi}{6}$, giving the fraction

$$\frac{5\pi/6}{2\pi} = \boxed{\frac{5}{12}}.$$

The correct answer is B.

Q16

Trigonometric Sequences

Answer B

Question 16

Find

$$\sum_{r=0}^{100} \cos\left(\frac{2r+1}{4}\pi\right).$$

A 0	B $\frac{\sqrt{2}}{2}$
C $\sqrt{2}$	D $2\sqrt{2}$
E $100\sqrt{2}$	F $101\sqrt{2}$

WORKED SOLUTION

The cosine values repeat every four terms:

$$\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2},$$

and each block sums to 0. There are $101 = 25 \cdot 4 + 1$ terms, so only the first term of the next block remains:

$$\boxed{\frac{\sqrt{2}}{2}}.$$

The correct answer is B.

Q17

Trigonometric Equations

Answer **E**

Question 17

Find the number of solutions of

$$\sin 5x = \cos 2x$$

in $0 \le x \le 2\pi$.

A 5	**B** 6
C 7	**D** 8
E 9	**F** 10

WORKED SOLUTION

Write

$$\cos 2x = \sin\left(\frac{\pi}{2} - 2x\right).$$

For $\sin A = \sin B$, either $A = B + 2k\pi$, or $A = \pi - B + 2k\pi$.

The first family gives

$$5x = \frac{\pi}{2} - 2x + 2k\pi \quad \Rightarrow \quad x = \frac{(4k+1)\pi}{14}.$$

Within $0 \le x \le 2\pi$, this gives $k = 0, 1, \ldots, 6$, so there are seven values.

The second family gives

$$5x = \pi - \left(\frac{\pi}{2} - 2x\right) + 2k\pi \quad \Rightarrow \quad x = \frac{(4k+1)\pi}{6}.$$

Within the interval, this gives $k = 0, 1, 2$, so there are three values.

The value $x = \frac{3\pi}{2}$ occurs in both families, and it is the only overlap. Therefore the number of distinct solutions is

$$7 + 3 - 1 = \boxed{9}.$$

The correct answer is E.

Q18

Trigonometric Equations

Answer **A**

Question 18

The number x satisfies

$$\sqrt{2}\sin 3x - 2\cos 3x = 1 - \sqrt{2},$$

$$\sin 3x + 2\sqrt{2}\cos 3x = \frac{1}{2}(4 + \sqrt{2}),$$

where $-180^\circ \le x \le 180^\circ$. Find the sum of all possible values of x.

- **A** 45°
- **B** 90°
- **C** 135°
- **D** 180°
- **E** 225°
- **F** 0°

WORKED SOLUTION

Let $s = \sin 3x$ and $c = \cos 3x$. Multiply the first equation by $\sqrt{2}$:

$$2s - 2\sqrt{2}c = \sqrt{2} - 2.$$

Multiply the second equation by 2:

$$2s + 4\sqrt{2}c = 4 + \sqrt{2}.$$

Subtracting gives $6\sqrt{2}c = 6$, so $c = \frac{\sqrt{2}}{2}$. Substitution gives $s = \frac{\sqrt{2}}{2}$. Therefore

$$3x = 45^\circ + 360^\circ k, \qquad x = 15^\circ + 120^\circ k.$$

Within the stated interval,

$$x = -105^\circ,\ 15^\circ,\ 135^\circ.$$

Their sum is $\boxed{45^\circ}$.

The correct answer is A.

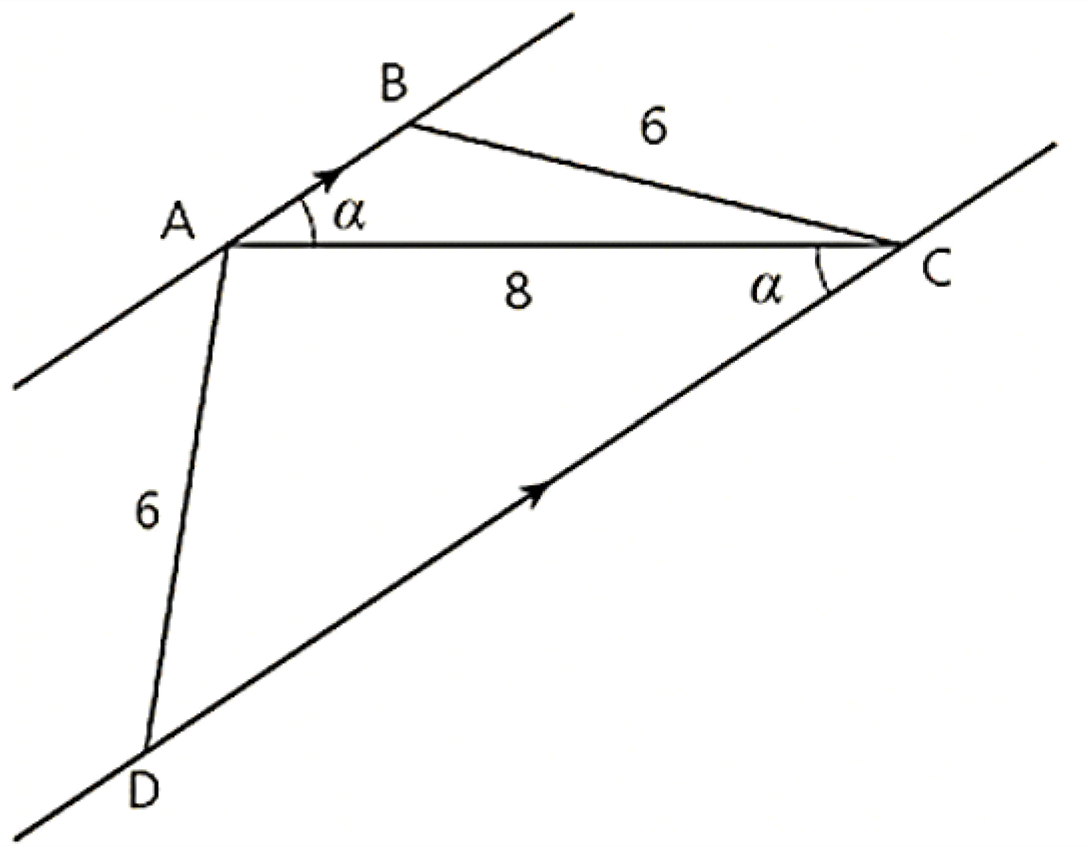
Q19

Trigonometric Geometry

Answer C

Question 19

In trapezium $ABCD$, $AB \parallel CD$, $AD = BC = 6$, and $AC = 8$. The diagonal AC makes an angle α with both parallel sides, as shown.



The area of triangle ACD is five times the area of triangle ABC. Find $\cos\alpha$.

A $\frac{\sqrt{14}}{3}$

B $\frac{3}{4}$

C $\frac{3\sqrt{35}}{20}$

D $\frac{\sqrt{42}}{7}$

E $\frac{7\sqrt{42}}{48}$

F $\frac{\sqrt{28}}{6}$

WORKED SOLUTION

The two triangles have the same altitude factor relative to AC, so

$$\frac{[ACD]}{[ABC]} = \frac{CD}{AB} = 5.$$

Let $AB = b$. Then $CD = 5b$.

Applying the cosine rule in triangle ABC,

$$6^2 = b^2 + 8^2 - 16b\cos\alpha,$$

so

$$\cos\alpha = \frac{b^2+28}{16b}.$$

In triangle ACD,

$$6^2 = (5b)^2 + 8^2 - 80b\cos\alpha,$$

so

$$\cos\alpha = \frac{25b^2+28}{80b}.$$

Equating these expressions gives

$$5(b^2+28) = 25b^2+28 \implies b^2 = \frac{28}{5}.$$

Substitution yields

$$\cos\alpha = \boxed{\frac{3\sqrt{35}}{20}}.$$

The correct answer is C.

Q20

Trigonometric Graphs

Answer D

Question 20

Find the maximum value of

$$\left(\frac{1}{2-\sin(x+135^\circ)}\right)^2$$

for $90^\circ \le x \le 270^\circ$.

A $\frac{1}{4}$

B $\frac{1}{9}$

C 2

D $\frac{18+8\sqrt{2}}{49}$

E 4

WORKED SOLUTION

Put $t = x + 135^\circ$. Then

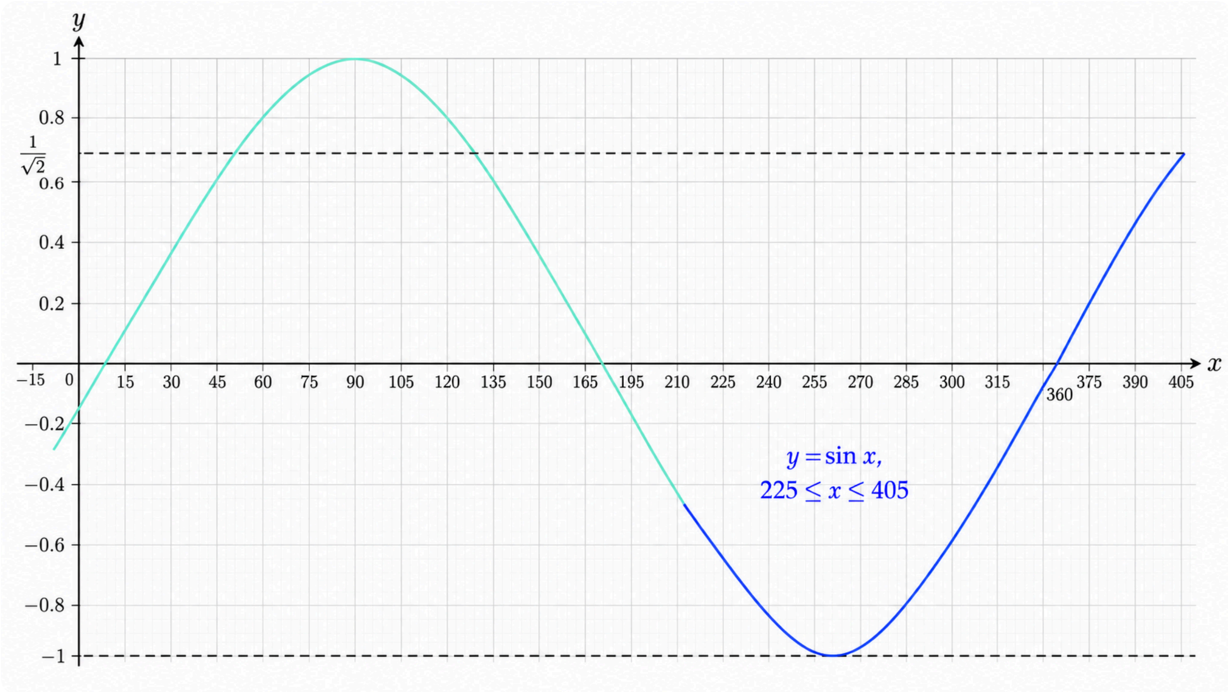
$$225^\circ \le t \le 405^\circ.$$

The expression is largest when $2 - \sin t$ is smallest, so we maximise $\sin t$ over this interval. Its maximum occurs at the endpoint $t = 405^\circ$, where

$$\sin 405^\circ = \frac{\sqrt{2}}{2}.$$

Therefore the maximum is

$$\frac{1}{\left(2-\frac{\sqrt{2}}{2}\right)^2} = \boxed{\frac{18+8\sqrt{2}}{49}}.$$



The correct answer is D.